



KOM4531 ADVANCED INDUSTRIAL AUTOMATION SYSTEMS APPLICATIONS

Servo Motors and Drives

FEATURES OF SERVO MOTORS AND DRIVES



- Servomotors excel at position and speed control.
- Precise and flexible positioning is possible.
- Servomotors do not stall even at high speeds. Deviations due to large external forces are corrected because encoders are used to monitor movement.

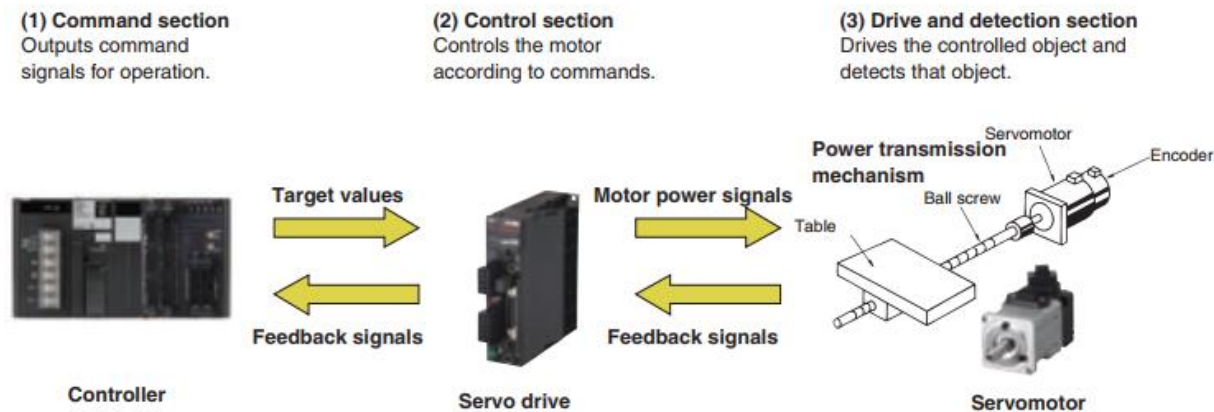


I576-E1-07

WHAT IS A SERVOMOTOR AND WHAT IS A SERVO DRIVE?



- A servomotor is a structural unit of a servo system and is used with a servo drive. The servomotor includes the motor that drives the load and a position detection component, such as an encoder.
- The servo system vary the controlled amount, such as position, speed, or torque, according to the set target value (command value) to precisely control the machine operation.



SERVOMOTOR

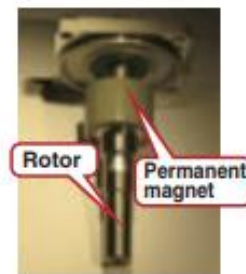
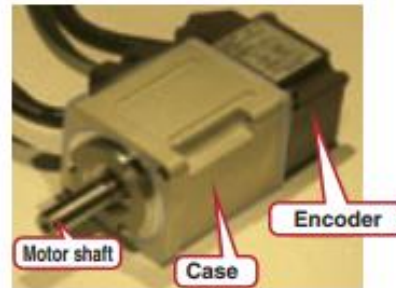


- The most common types of industrial servomotors are those based on brushless motors.
- The rotor has a powerful permanent magnet. The stator is composed of multiple conductor coils, and the rotor spins when the coils are powered in the specified order.
- The movement of the rotor is determined by the stator's frequency, phase, polarity, and current when the correct current is supplied to the stator coils at the appropriate time.

WHAT IS A SERVOMOTOR AND WHAT IS A SERVO DRIVE?



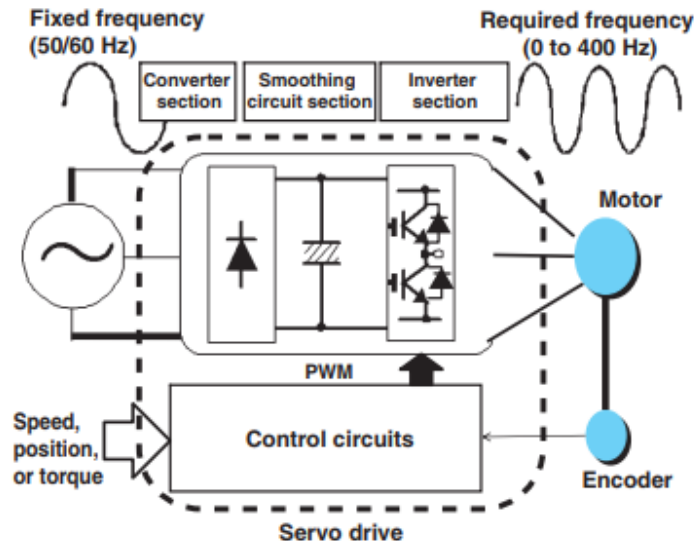
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WHAT IS A SERVOMOTOR AND WHAT IS A SERVO DRIVE?



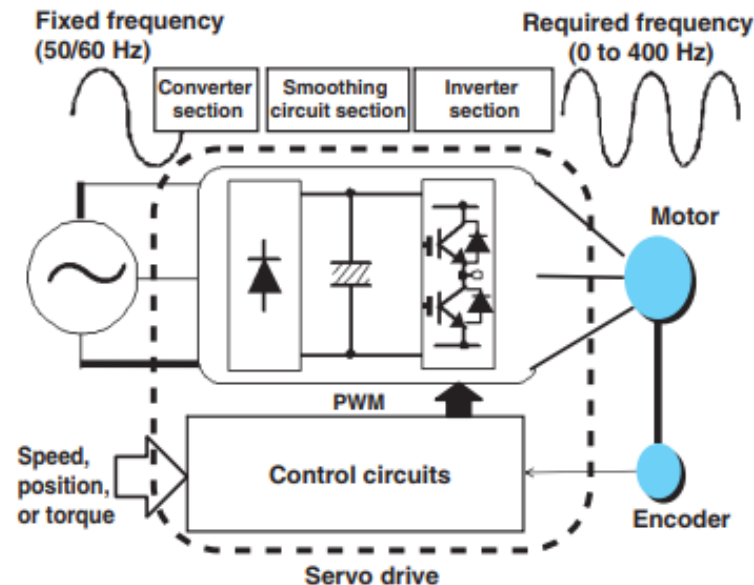
- **Servo drives** follow commands from the host controller and control the output torque, rotation speed, or position of motors. The position, speed, or torque are controlled according to inputs from a motion controller, feedback encoder, and the servomotor itself, and the servo drive supplies the appropriate amounts of power to the servomotor at the appropriate times.
- The basic operating principle is the same as for an inverter, in which the motor is operated by converting AC power to DC power to be a certain frequency.



A SERVO DRIVE FUNCTIONS



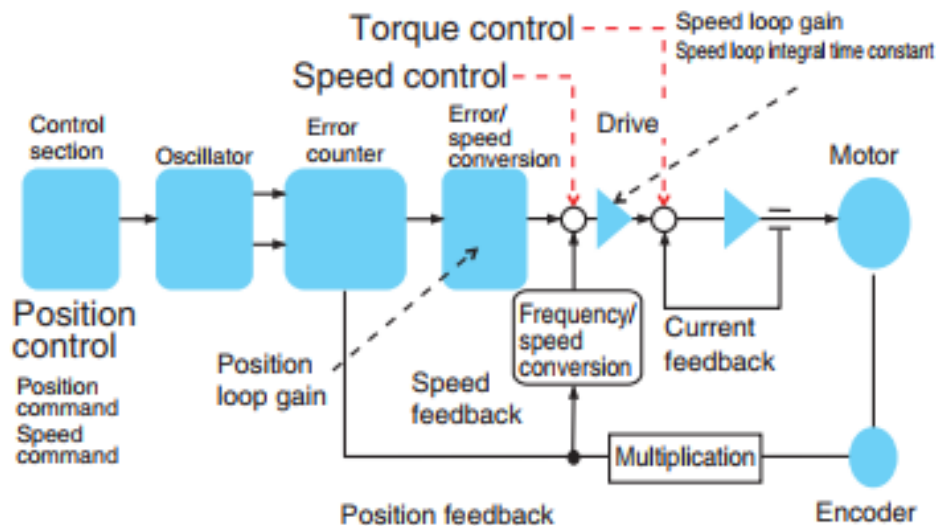
- Communications with the motion controller
- Encoder feedback reading and realtime closed-loop control adjustment
- I/O processing for safety components, mode inputs, and operating status output signals



SERVO OPERATION AND CONFIGURATION



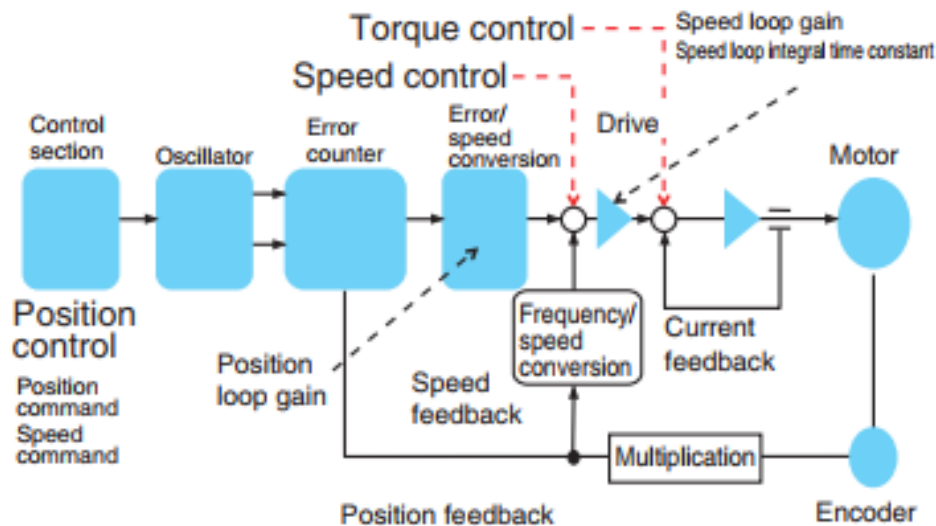
- A system built with servo drives and servomotors controls motor operation in closed loop.
- The actual position, speed, or torque of the servomotor is fed back to compare to the command value and calculate the following errors between them.
- The servo drive corrects the operation of the servomotor in realtime using this error information to ensure that the system can achieve the required performance.



SERVO OPERATION AND CONFIGURATION CONT'D



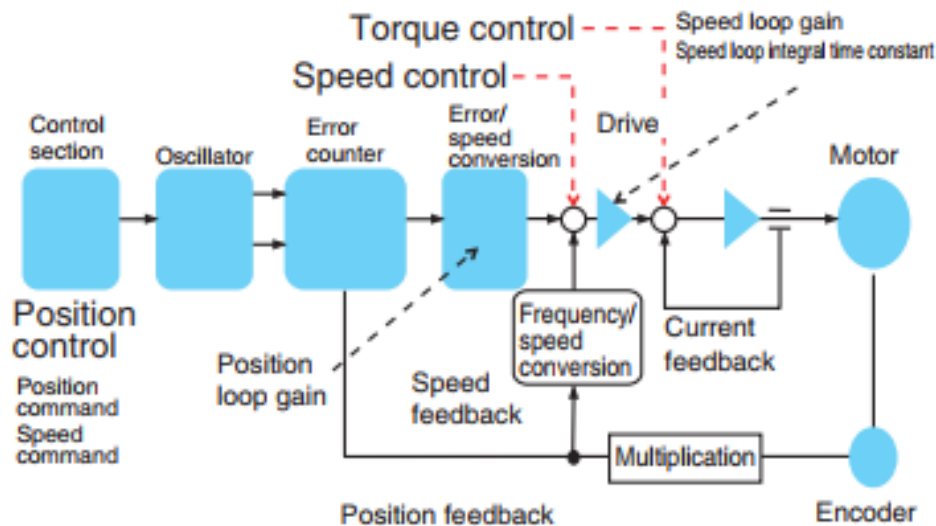
- This cycle of feedback, error detection, and correction is called closed-loop control. The control loop is processed by either of servo drive or motion controller, or both depending on the required control. The control loops for position, speed, and torque are independently used to achieve the required operation.
- Applications will not always require all three control loops. In some applications, only the control loop for torque control will be required. In other applications, current and speed for speed control are required, and in still other applications, three control loops for position control are required.



CONTROL LOOP



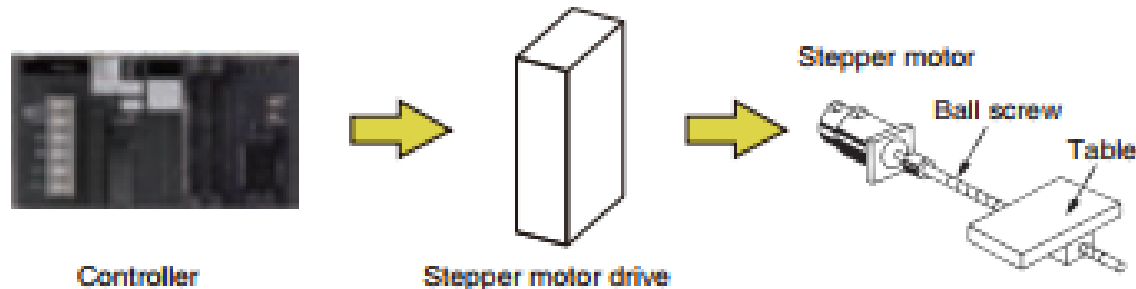
- The **control loop** is processed by either of servo drive or motion controller, or both depending on the required control. The control loops for position, speed, and torque are independently used to achieve the required operation. Applications will not always require all three control loops (. In some applications, only the control loop for torque control will be required. In other applications, current and speed for speed control are required, and in still other applications, three control loops for position control are required



OPEN LOOP VS CLOSED LOOP



- **In Open Loop**, A stepper motor is used instead of a servomotor. There is no feedback loop. The overall configuration is simple. Positioning can be performed at low cost, but gear and ball screw backlash and pitch errors cannot be compensated. When a stepper motor stalls, an error will occur between the command value and the actual movement. This error cannot be compensated.
- Open loop control is suitable for low-precision, low-cost, low-speed, and low-load-change applications.



OPEN LOOP VS CLOSED LOOP



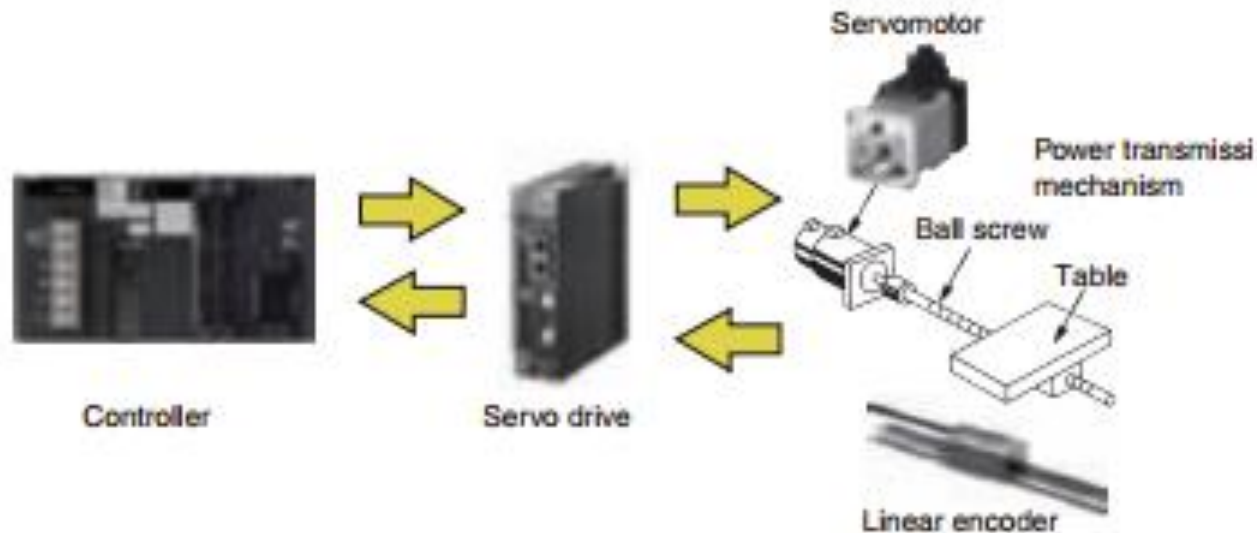
- **In Semi-Closed Loop**, Typically an encoder or other detector is attached behind the motor. The encoder detects the rotation angle of a feed screw (ball screw) and provides it as feedback of the machine (workpiece or table) travel position. This means that the position of the machine is not detected directly.



OPEN LOOP VS CLOSED LOOP



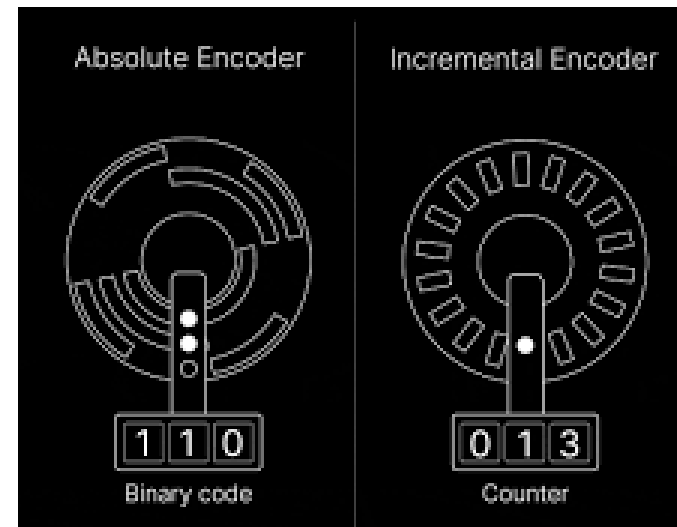
- **In Closed Loop**, The motor is controlled while directly reading the position of the machine (workpiece or table) using a linear encoder and comparing the read position with the command value (target value). Therefore, there is no need to compensate for gear backlash between the motor and mechanical system, feed screw pitch error, or error due to feed screw torsion or expansion



ENCODERS



- **Encoders** are one of the hardware elements that form the core of a servo system, and they generate speed and position feedback.
- In many cases, the encoder is built into the servomotor or attached to the servomotor. In certain applications, the encoder is an independent unit that is installed away from the servomotor. When the encoder is installed in a remote location, it is used for related parameters in addition to for control of servomotor operation.
- Encoders are divided into two kinds:
 - Absolute Encoder
 - Incremental Encoder



ENCODERS

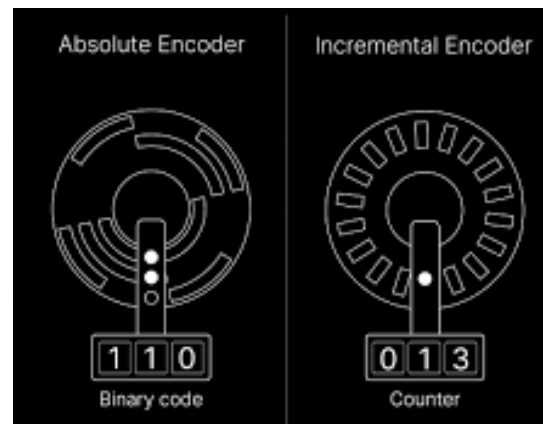


- **Absolute Encoder**

- An absolute encoder provides a unique digital code for each position
- The position data is retained even if power is lost, so it resumes from where it left off when restarted
- Therefore, each position is “absolute” and specific.

- **Incremental Encoder**

- An incremental encoder determines the position relative to a starting point using increasing or decreasing counts
- If power is interrupted, the position data is lost, and the system may need to return to a reference position.

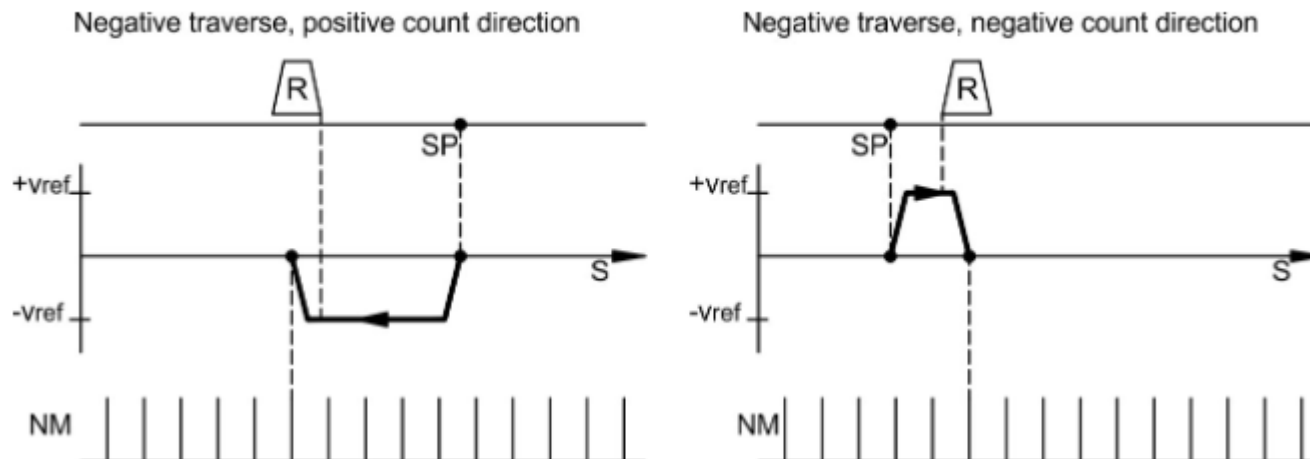


HOMING



Sets the reference point to the setpoint position (the following error is lost). The load does not move.

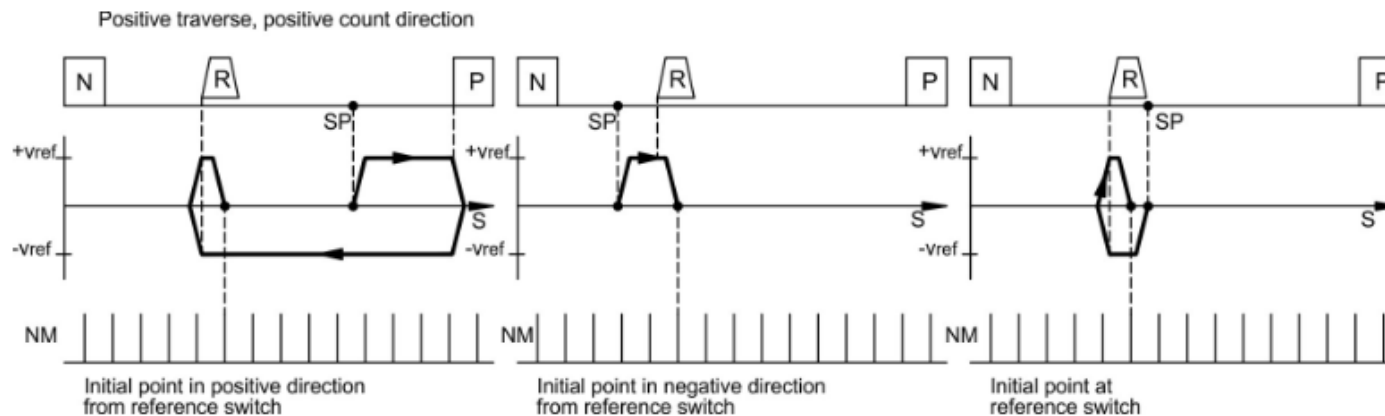
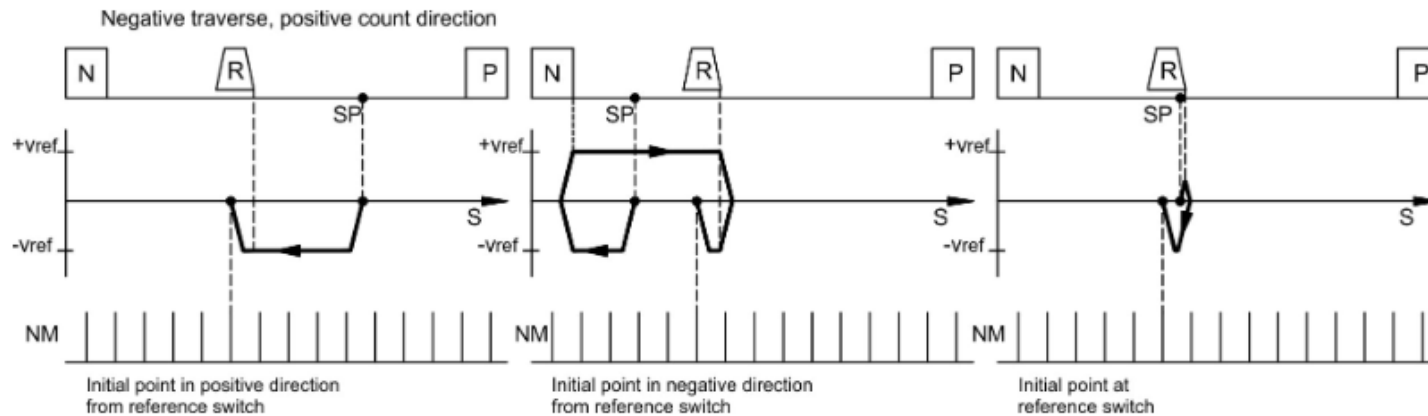
Homing without limit switches, with homing switch, with feedback zero



HOMING



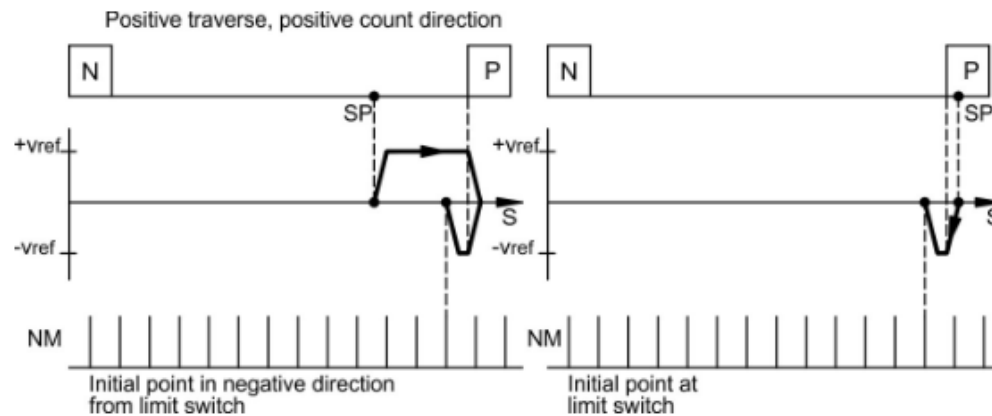
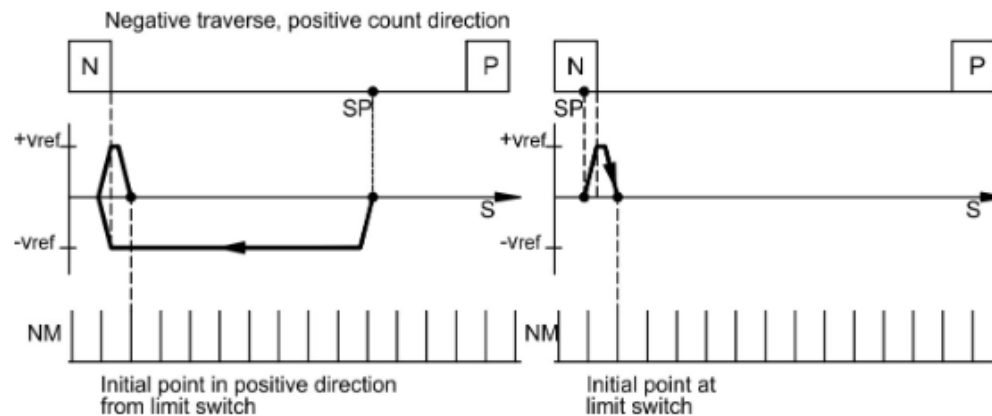
Homing with limit switches, with homing switch, with feedback zero



HOMING



Homing with limit switches, without homing switch, with feedback zero

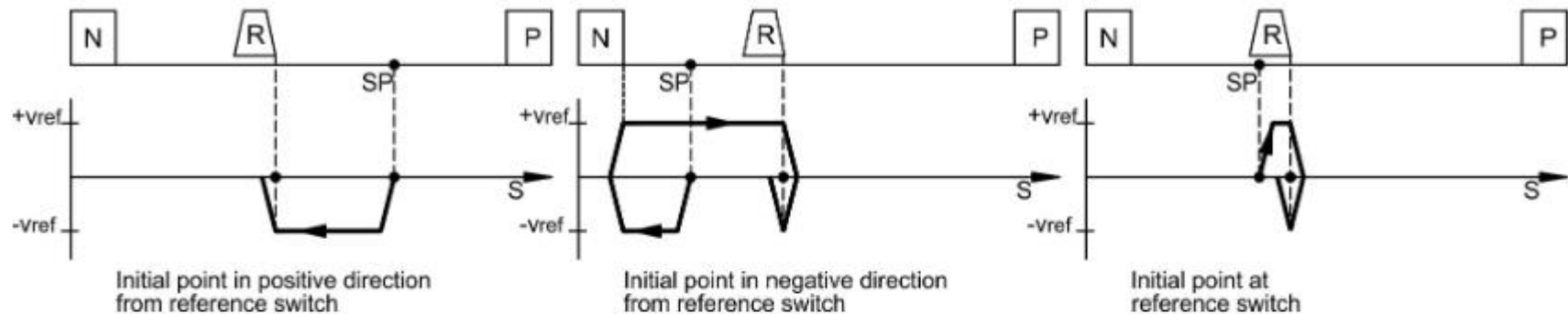


HOMING

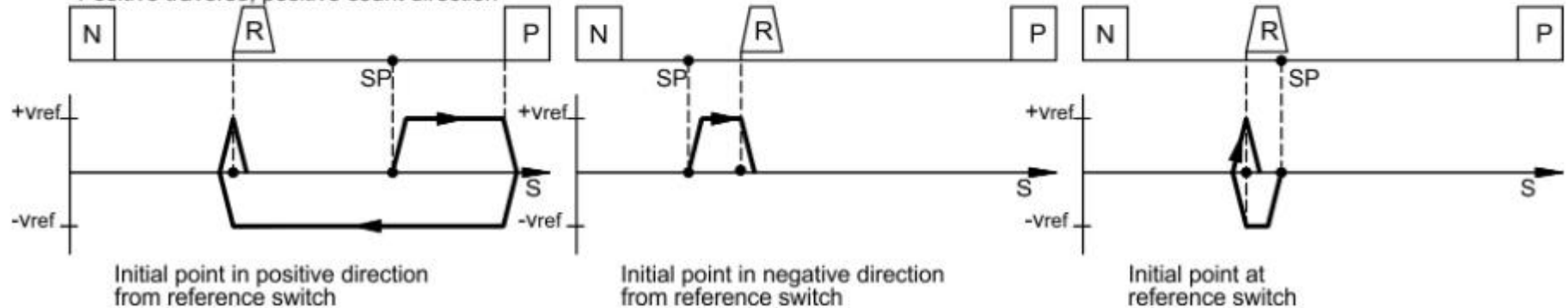


Homing with limit switches, with homing switch, without feedback zero

Negative traverse, positive count direction



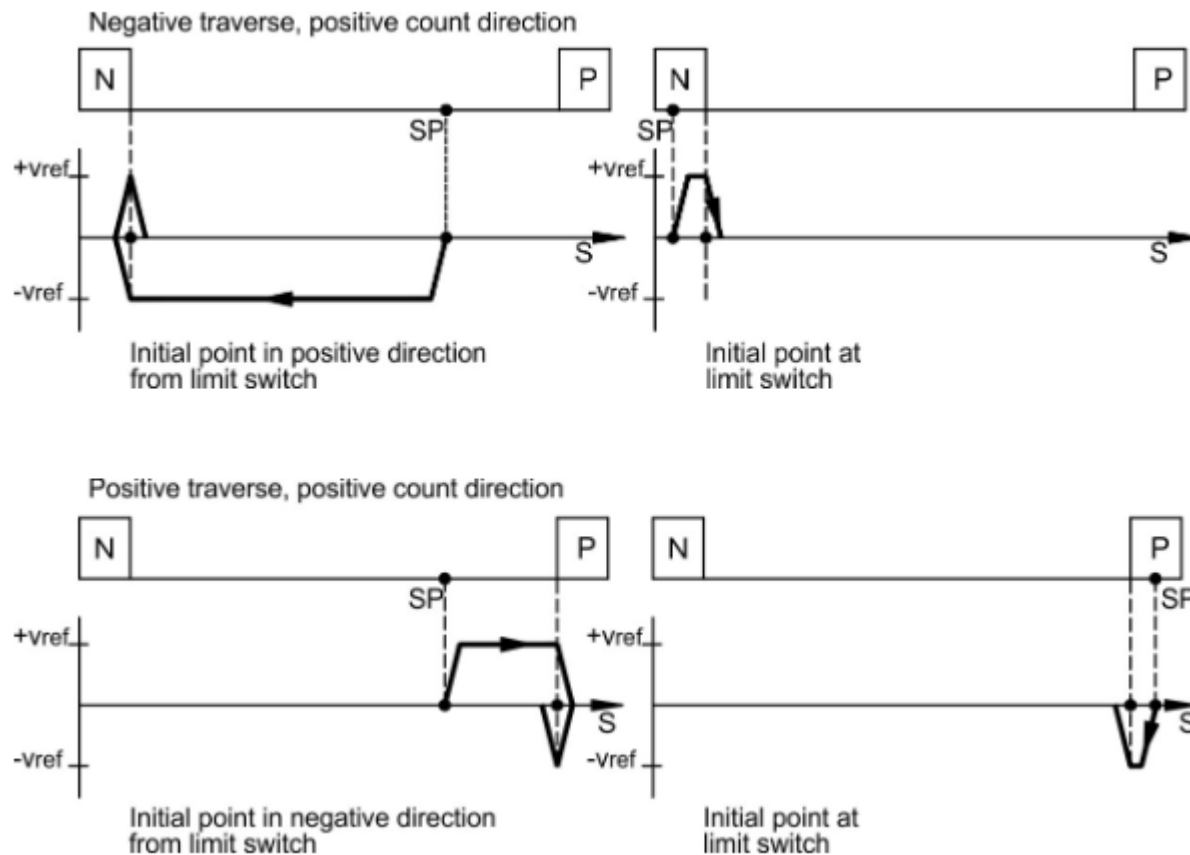
Positive traverse, positive count direction



HOMING



Homing with limit switches, without homing switch, without feedback zero



EXPLANATION OF TERMS



There are numerous items that effect the **performance** of servomotors and servo drives. Some of them are listed below:

Effective Torque

A value of the average torque (RMS) that is produced during operation of a motor. A motor with a larger value than the effective torque must be chosen.

The unit is $\text{N}\cdot\text{m}$.

Torque Constant

When a current flows to a motor, the current and the flux produce a torque. The torque constant is the relationship between this current and the produced torque. The higher the torque, the smaller the controlling current.

The unit is $\text{N}\cdot\text{m}/\text{A}$.

Power Rate

The power rate is given by this formula:

Power rate = (Rated torque) 2 /Rotor inertia $\times 10^{-3}$. The higher the value is, the better the response is. The unit is kW/s .

Rotor Inertia

The moment of inertia of the rotor, expressed in Jm . The smaller the value is, the quicker the response is. The unit is

$\text{kg}\cdot\text{m}^2$.

Applicable Load Inertia

The range in which a drive can control the load inertia. The range is limited by the gain adjustment range and the energy absorption capacity. The unit is $\text{kg}\cdot\text{m}^2$.

Rated Output

The rated output (P) is the mechanical power that a motor can output. The rated torque (T) and the rated speed (N) are related to the rated power as follows:

$$P=0.105\times T\times N$$

Backlash

The mechanical system has a dead zone between forward and reverse. A gear that changes from forward to reverse must turn by the amount of the dead zone before turning the specified amount. This movement is called the backlash.

Backlash is given in minutes. One turn is 360 degrees. One minute is $1/60$ of 1 degree.

The smaller the backlash is, the less the dead zone is.

EXPLANATION OF TERMS



Like performance terms, there are also bunch of **functions**. Some of them are listed as follows :

Realtime Autotuning

Realtime autotuning estimates the load inertia of the machine in realtime, and operates the machine by automatically setting the gain according to the estimated load inertia. At the same time, it can lower the resonance and vibration if the adaptive filter is enabled.

Manual Tuning

A gain adjustment method used when autotuning cannot be performed due to the restrictions of the operating pattern or load conditions or when maximum responsiveness needs to be obtained for individual loads.

Notch Filter

A notch filter is used to eliminate a specified frequency component. The notch filter can restrict a resonance peak, and it allows a high gain setting and vibration reduction.

Feed-forward Function

A function that increases the responsiveness of the control system by adding the feed-forward value to the command value.

Instantaneous Speed Observer Function

This function uses a load model to estimate the motor speed. It improves the speed detection accuracy and can provide both high responsiveness and minimum vibration when stopping.

Safe Torque OFF Function

The safe torque OFF function (STO) is used to cut off the motor current and stop the motor through the input signals from a safety device, such as a safety controller or safety sensor.

Damping Control

A function used to reduce vibration when using a low-rigidity mechanism or equipment whose ends tend to vibrate.

Internally Set Speed Control

A function that controls the speed of the servomotor using speeds set in the internal speed setting parameters.

Free Run

A status in which a motor continues to rotate due to its inertia when servo is turned OFF.

EXPLANATION OF TERMS



Like performance terms, there are also bunch of **functions**. Some of them are listed as follows :

Electronic Gear

A function that rotates the servomotor for the number of pulses obtained by multiplying the command pulses by the electronic gear ratio. The electronic gear is used to synchronize the position and speed of two lines, to enable using a position controller with a low command pulse frequency or to set the machine travel distance per pulse, to 0.01 mm for example.

Torque Limit

A function that limits the output torque of a motor. The torque limit is used for pressing a moving part of a machine (such as a bending machine) against a workpiece with a constant force, or for protecting the servomotor and mechanical system from excessive force or torque.

Position Command Filter

A function that performs soft start processing for the command pulses using the selected filter to gently accelerate and decelerate. The filter characteristics for the position command filter are selected using the Position Command Filter Time Constant Setting.

This function is effective when there is no acceleration or

deceleration function in the command pulse (controller), when the command pulse frequency changes abruptly, causing the machinery to vibrate during acceleration and deceleration, or when the electronic gear setting is high.

Position Loop Gain

Servo systems with a low loop gain have a low response and can increase the positioning time.

The higher the position loop gain, the shorter the positioning time. If the setting is too high, however, overshooting or hunting may occur in the system.

Incremental Command

An incremental command determines the travel amount between the present position and the target position.

Absolute Command

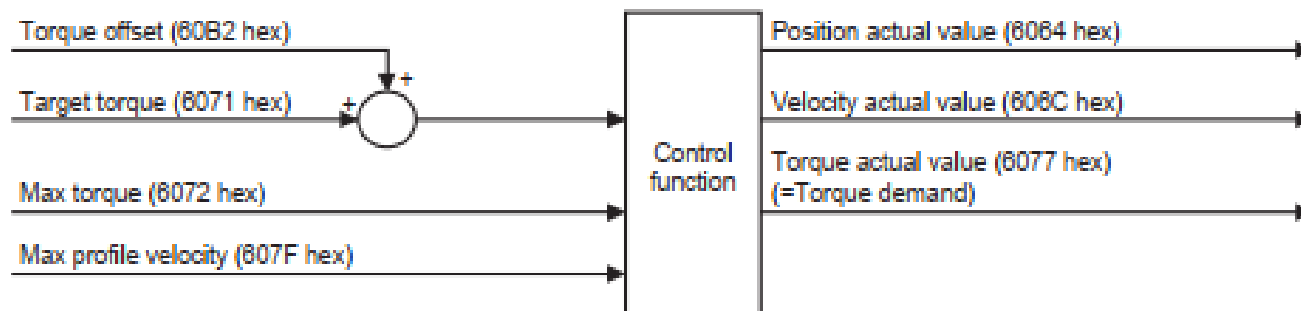
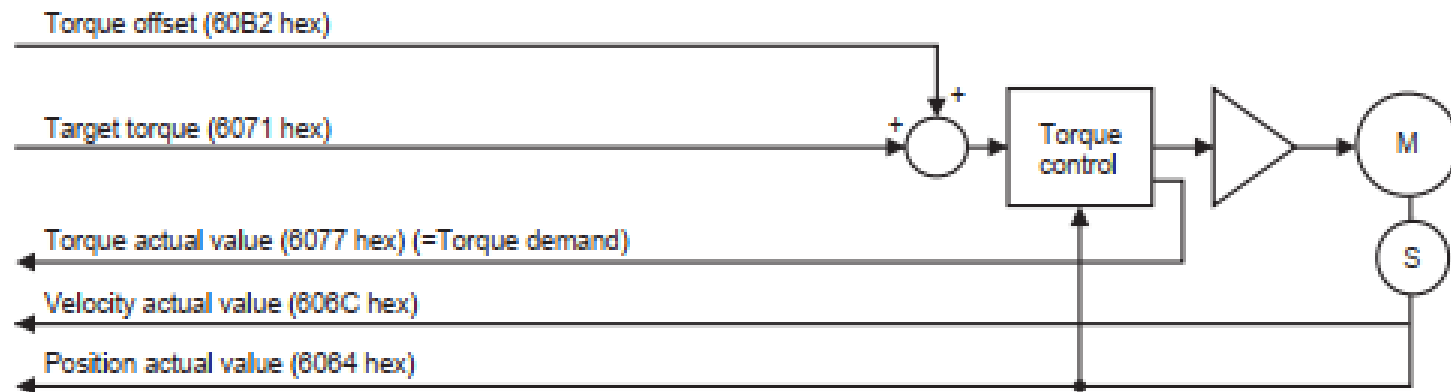
An absolute command determines the travel amount from a command value that is based upon the origin. Thus, the command value is different from the travel amount unless the motor is at the origin.

CONTROL FUNCTIONS



- There are variations of control functions :
 - **Cyclic Synchronous Torque Mode:** In this mode of operation, the controller has a path generation function (an operation profile calculation function) and it gives the target torque to the Servo Drive using cyclic synchronization. Torque control is performed by the Servo Drive.
 - **Cyclic Synchronous Velocity Mode:** In this mode of operation, the controller has a path generation function (an operation profile calculation function) and it gives the target speed to the Servo Drive using cyclic synchronization. Speed control and torque control are performed by the Servo Drive.
 - **Cyclic Synchronous Position Mode:** In this mode of operation, the controller has a path generation function (an operation profile calculation function) and it gives the target position to the Servo Drive using cyclic synchronization. Position control, speed control, and torque control are performed by the Servo Drive.

CYCLIC SYNCHRONOUS TORQUE MODE

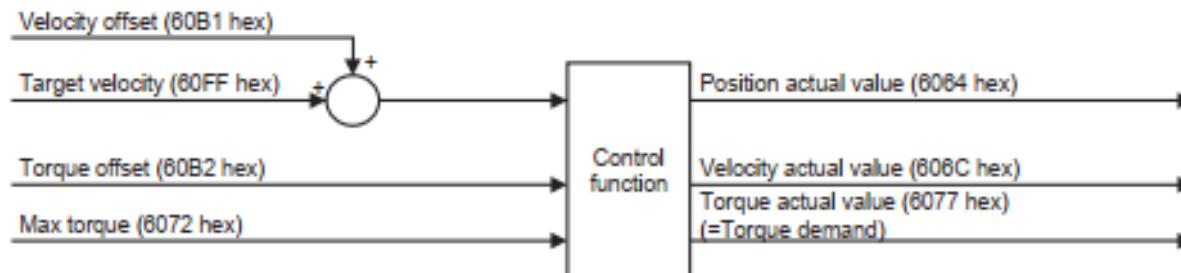
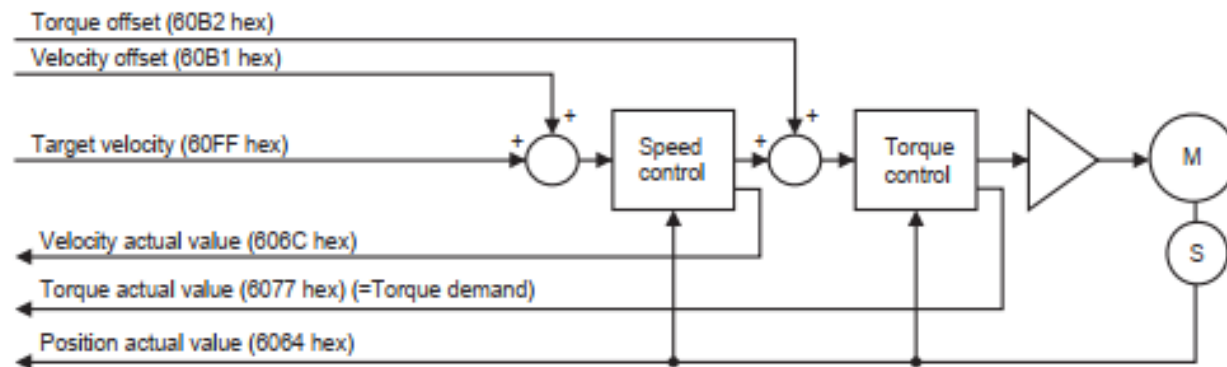


CYCLIC SYNCHRONOUS TORQUE MODE



Index	Sub-index	Name	Access	Size	Unit	Setting range	Default setting
6040 hex	00 hex	Controlword	RW	U16	0 to FFFF hex	–	0000 hex
6060 hex	00 hex	Modes of operation	RW	INT8	–	0 to 10	0
6071 hex	00 hex	Target torque	RW	INT16	0.1%	–5,000 to 5,000	0000 hex
6072 hex	00 hex	Max torque	RW	U16	0.1%	0 to 5,000	5000
607F hex	00 hex	Max profile velocity	RW	U32	Command units/s	0 to 2,147,483,647	0000 hex
60B2 hex	00 hex	Torque offset	RW	INT16	0.1%	–5,000 to 5,000	0
6041 hex	00 hex	Statusword	RO	U16	0 to FFFF hex	–	0000 hex
6064 hex	00 hex	Position actual value	RO	INT32	Command units	–2,147,483,648 to 2,147,483,647	0000 hex
606C hex	00 hex	Velocity actual value	RO	INT32	Command units/s	–2,147,483,647 to 2,147,483,647	0000 hex
6077 hex	00 hex	Torque actual value	RO	INT16	0.1%	–5,000 to 5,000	0000 hex

CYCLIC SYNCHRONOUS VELOCITY MODE

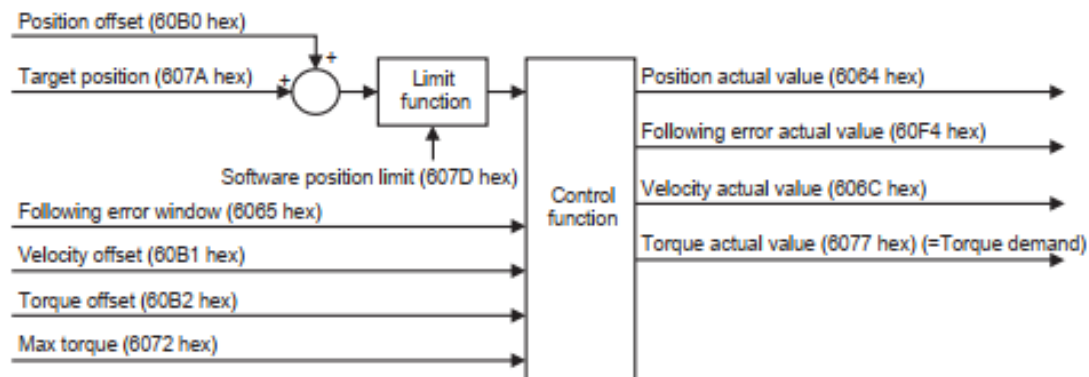
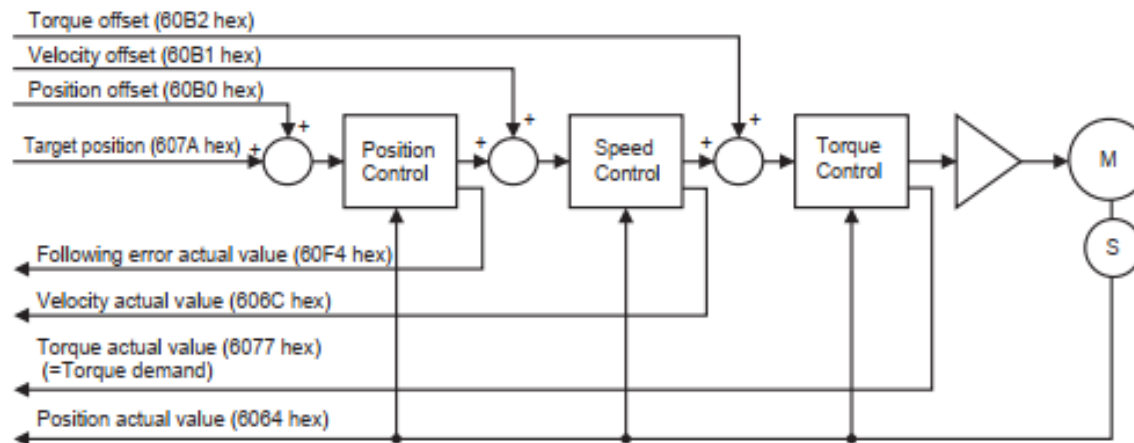


CYCLIC SYNCHRONOUS VELOCITY MODE



Index	Sub-index	Name	Access	Size	Unit	Setting range	Default setting
6040 hex	00 hex	Controlword	RW	U16	0 to FFFF hex	–	0000 hex
6060 hex	00 hex	Modes of operation	RW	INT8	–	0 to 10	0
60FF hex	00 hex	Target velocity	RW	INT32	Command units/s	–2,147,483,647 to 2,147,483,647	0000 hex
6072 hex	00 hex	Max torque	RW	U16	0.1%	0 to 5,000	5000
60B1 hex	00 hex	Velocity offset	RW	INT32	Command units/s	–2,147,483,648 to 2,147,483,647	0000 hex
60B2 hex	00 hex	Torque offset	RW	INT16	0.1%	–5,000 to 5,000	0
6041 hex	00 hex	Statusword	RO	U16	0 to FFFF hex	–	0000 hex
6064 hex	00 hex	Position actual value	RO	INT32	Command units	–2,147,483,648 to 2,147,483,647	0000 hex
606C hex	00 hex	Velocity actual value	RO	INT32	Command units/s	–2,147,483,647 to 2,147,483,647	0000 hex
6077 hex	00 hex	Torque actual value	RO	INT16	0.1%	–5,000 to 5,000	0000 hex

CYCLIC SYNCHRONOUS POSITION MODE



CYCLIC SYNCHRONOUS POSITION MODE

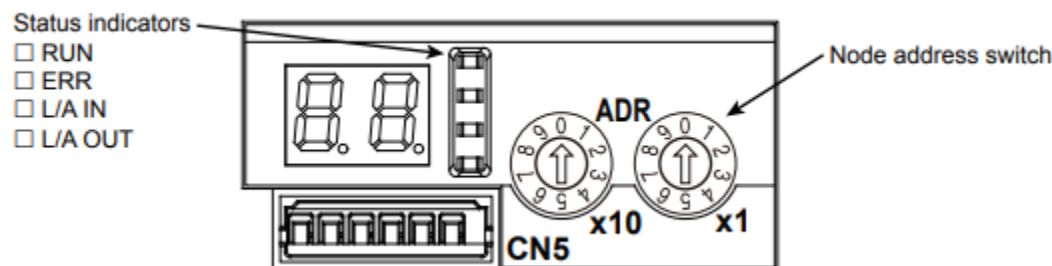


Index	Sub-index	Name	Access	Size	Unit	Setting range	Default setting
6040 hex	00 hex	Controlword	RW	U16	0 to FFFF hex	–	0000 hex
6060 hex	00 hex	Modes of operation	RW	INT8	–	0 to 10	0
607A hex	00 hex	Target position	RW	INT32	Command units	–2,147,483,648 to 2,147,483,647	0000 hex
6065 hex ^{*1}	00 hex	Following error window	RW	U32	Command units	0 to 134,217,728, or 4,294,967,295	100000
6072 hex	00 hex	Max torque	RW	U16	0.1%	0 to 5,000	5000
60B0 hex	00 hex	Position offset	RW	INT32	Command units	–2,147,483,648 to 2,147,483,647	0000 hex
60B1 hex	00 hex	Velocity offset	RW	INT32	Command units/s	–2,147,483,648 to 2,147,483,647	0000 hex
60B2 hex	00 hex	Torque offset	RW	INT16	0.1%	–5,000 to 5,000	0
6041 hex	00 hex	Statusword	RO	U16	0 to FFFF hex	–	0000 hex
6064 hex	00 hex	Position actual value	RO	INT32	Command units	–2,147,483,648 to 2,147,483,647	0000 hex
606C hex	00 hex	Velocity actual value	RO	INT32	Command units/s	–2,147,483,647 to 2,147,483,647	0000 hex
6077 hex	00 hex	Torque actual value	RO	INT16	0.1%	–5,000 to 5,000	0000 hex
60F4 hex	00 hex	Following error actual value	RO	INT32	Command units	–2,147,483,648 to 2,147,483,647	0000 hex

ETHERCAT COMMUNICATIONS



- Display area and settings are shown below:



- The node address switches located in the display area are used to set the EtherCAT node address.

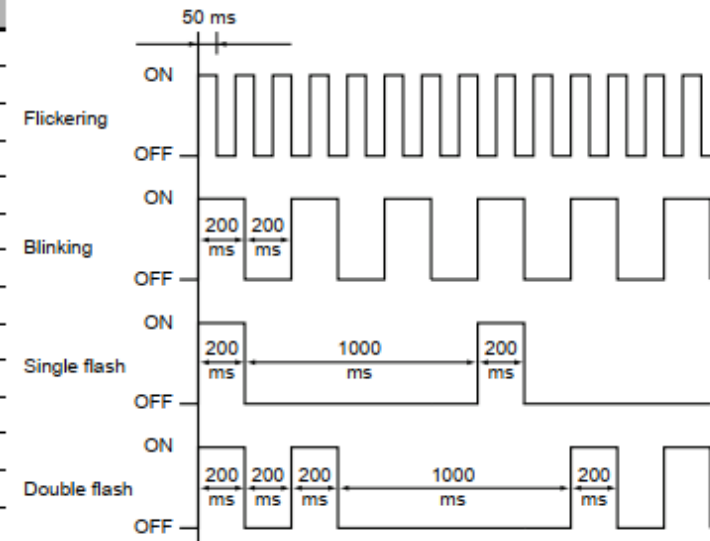
Node address switch setting	Description
	Connection to NJ/NX-series CPU Unit or CJ1W-NC□8□ Position Control Unit
00	The controller sets the node address.
01 to 99	The node address switch setting is used as the node address.

ETHERCAT COMMUNICATIONS

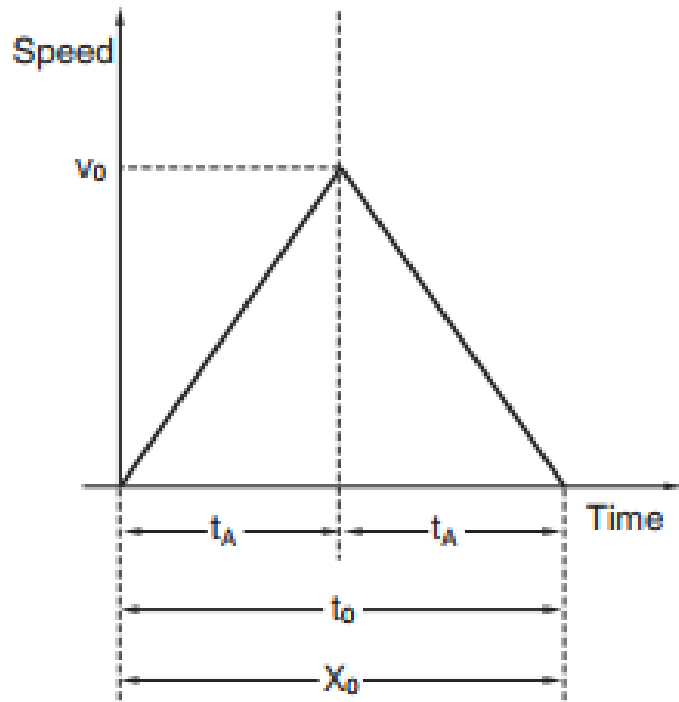


The following table shows the EtherCAT status indicators and their meaning.

Name	Color	Status	Description
RUN	Green	OFF	Init state
		Blinking	Pre-Operational state
		Single flash	Safe-Operational state
		ON	Operational state
ERR	Red	OFF	No error
		Blinking	Communications setting error
		Single flash	Synchronization error or communications data error
		Double flash	Application WDT timeout
		Flickering	Boot error
		ON	PDI WDT timeout
L/A IN	Green	OFF	Link not established in physical layer
		ON	Link established in physical layer
		Flickering	In operation after establishing link
L/A OUT	Green	OFF	Link not established in physical layer
		ON	Link established in physical layer
		Flickering	In operation after establishing link



OPERATING PATTERNS- TRIANGULAR



Maximum Speed $v_0 = \frac{X_0}{t_A}$ X_0 : Distance Moved in t_0 Time [mm]

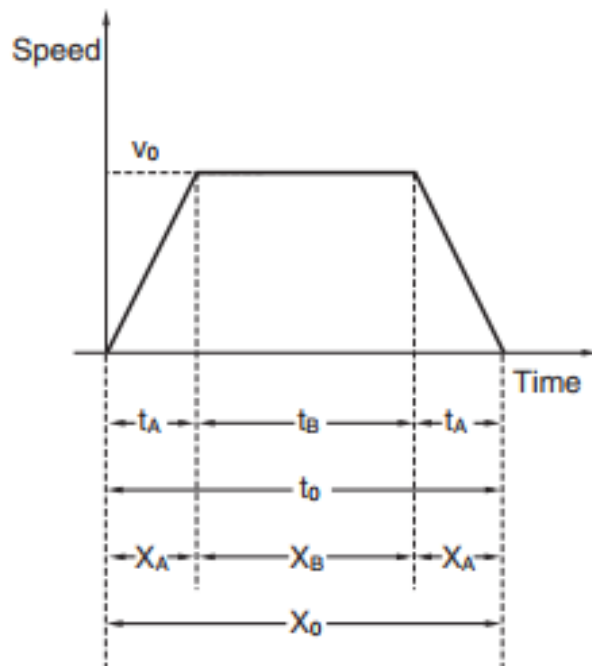
v_0 : Maximum Speed [mm/s]

Acceleration/Deceleration Time $t_A = \frac{X_0}{v_0}$ t_0 : Positioning Time [s]

t_A : Acceleration/
Deceleration Time [s]

Travel Distance $X_0 = v_0 \cdot t_A$

OPERATING PATTERNS- TRAPEZOID



Maximum Speed $v_0 = \frac{X_0}{t_0 - t_A}$

Acceleration/Deceleration Time $t_A = t_0 - \frac{X_0}{v_0}$

Total Travel Time $t_0 = t_A + \frac{X_0}{v_0}$

Constant-velocity travel time $t_B = t_0 - 2 \cdot t_A = \frac{2 \cdot X_0}{v_0} - t_0 = \frac{X_0}{v_0} - t_A$

Total Travel Distance $X_0 = v_0 (t_0 - t_A)$

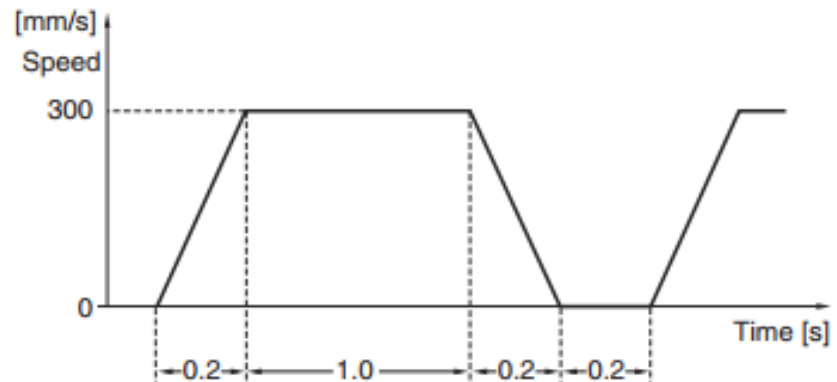
Acceleration/Deceleration Travel Distance $X_A = \frac{v_0 \cdot t_A}{2} = \frac{v_0 \cdot t_0 - X_0}{2}$

Constant-velocity travel distance $X_B = v_0 \cdot t_B = 2 \cdot X_0 - v_0 \cdot t_0$

EXAMPLE OF OPERATING PATTERN



- One Speed Change
- Velocity for a Load Travel $V = 300$ [mm/s]
- Strokes $L = 360$ [mm]
- Stroke Travel Time $t_S = 1.4$ [s]
- Acceleration/Deceleration Time $t_A = 0.2$ [s]
- Positioning Accuracy $A_p = 0.01$ [mm]

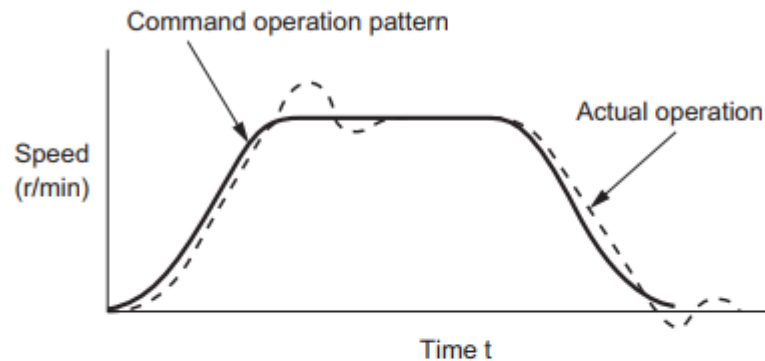


GAIN TUNING

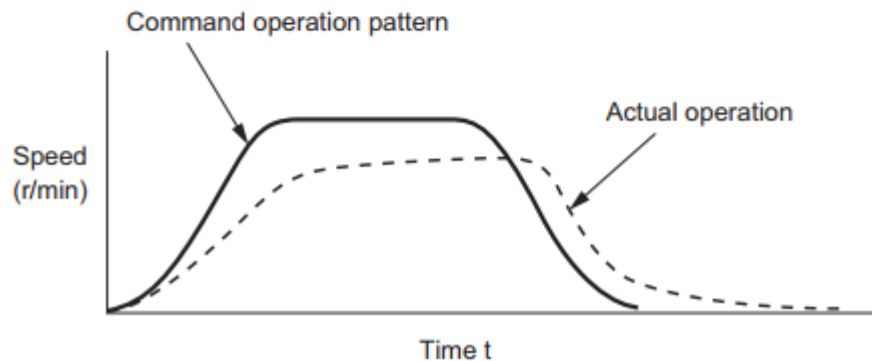


Response to the position loop gain adjustment is illustrated below.

- If the position loop gain is high, an overshooting occurs.



- If the position loop gain is low, positioning completion speed becomes slow.

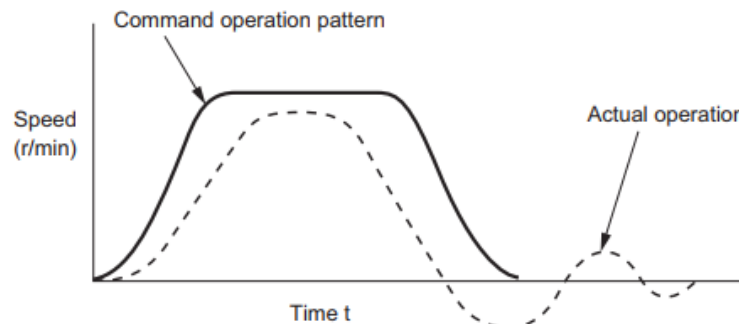


GAIN TUNING

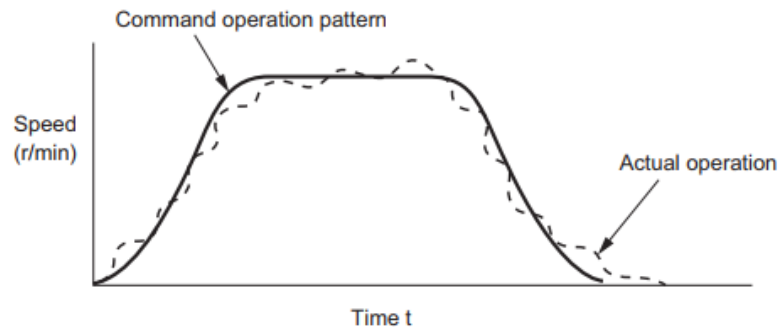


This is related to Speed Loop Integral Time Constant (3102 hex), and by increasing the integral time constant, the speed loop gain value can be increased.

- If the speed loop gain is low, the speed response becomes slow and a large overshooting occurs. In such case, increase the speed loop gain.



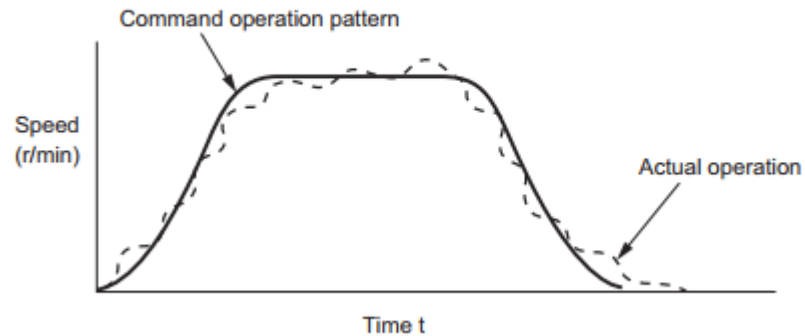
- If the speed loop gain is high, vibrations are more likely to occur. Vibration or resonance may not disappear. In such case, decrease the speed loop gain.



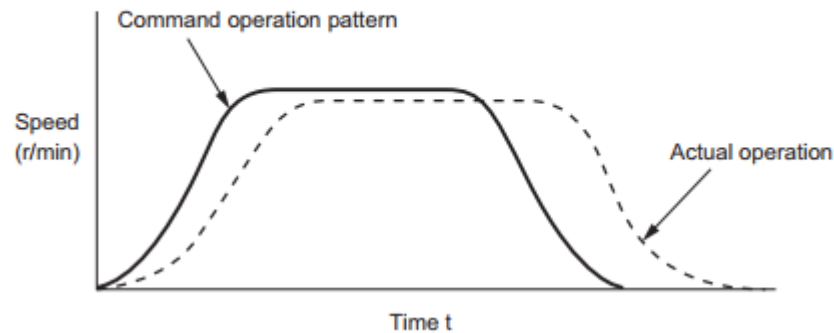
GAIN TUNING



- If the speed loop integral time constant is low, vibration or resonance occurs. In such case, increase the speed loop integral time constant.



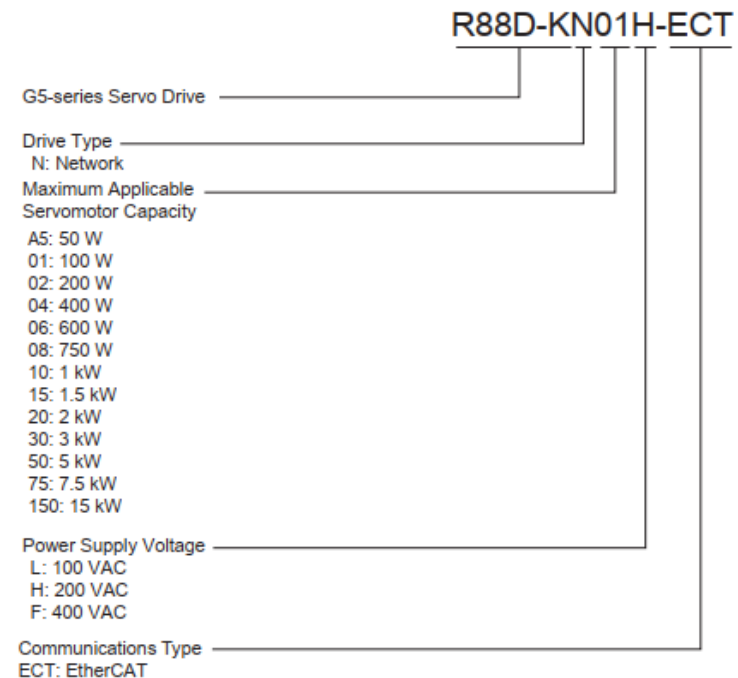
- If the speed loop integral time constant is high, the response is delayed. The servo rigidity becomes weak. In such case, decrease the speed loop integral time constant.



READING MODEL NUMBER- SERVO DRIVE



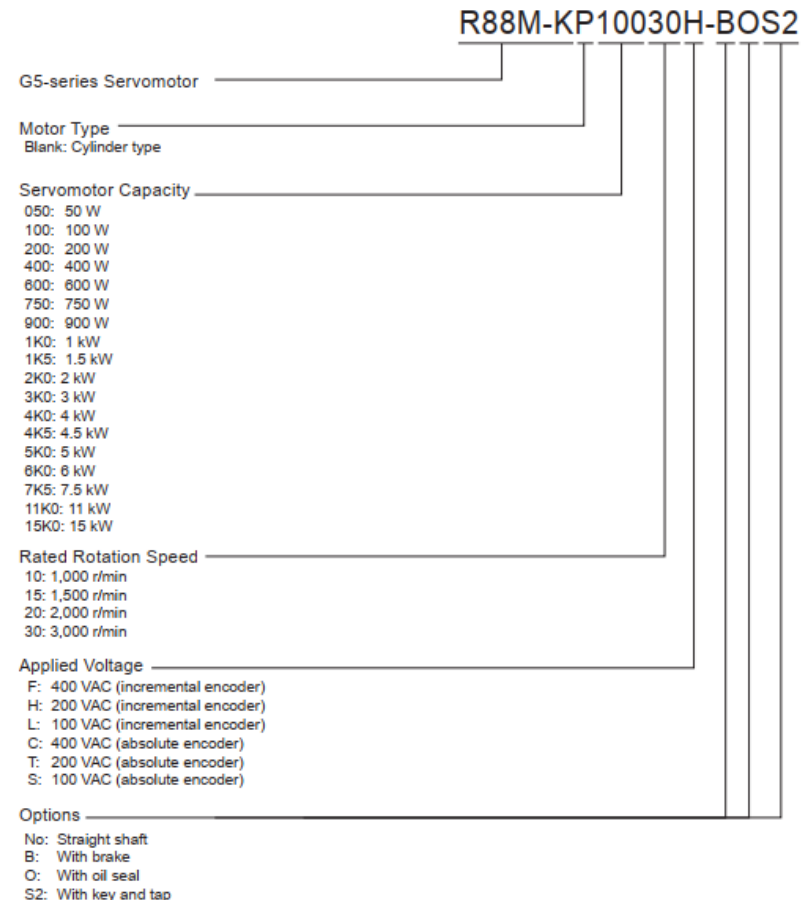
- The Servo Drive model number tells the Servo Drive type, applicable Servomotor capacity, power supply voltage, etc.



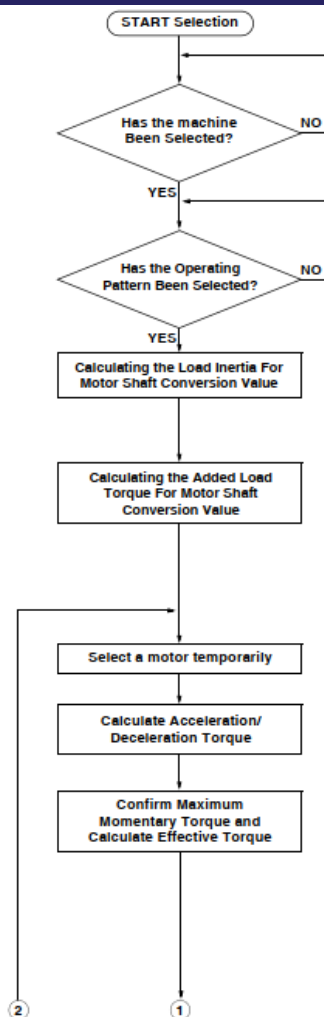
READING MODEL NUMBER- SERVOMOTOR



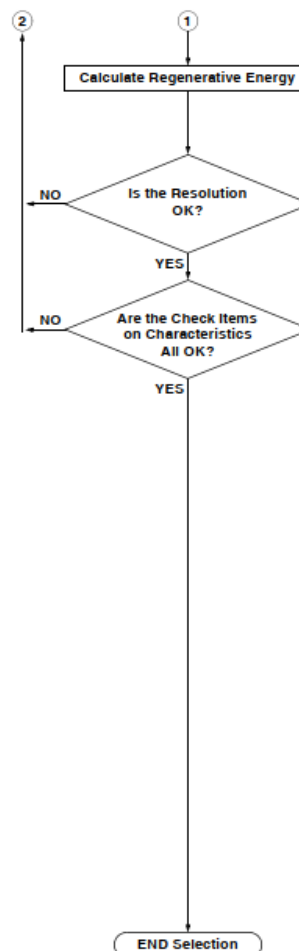
- The model number provides information such as the Servomotor type, motor capacity, rated rotation speed, and power supply voltage.



HOW TO SELECT THE SERVOMOTOR?



Explanation	References
Determine the size, mass, coefficient of friction, and external forces of all the moving part of the servomotor the rotation of which affects.	---
- Determine the operating pattern (relationship between time and speed) of each part that must be controlled. - Convert the operating pattern of each controlled element into the motor shaft operating pattern.	• Operation Pattern Formula
- The elements of the machine can be separated so that inertia can be calculated for each part that moves as the servomotor rotates. - Calculate the inertia applied to each element to calculate the total load inertia of the motor shaft conversion value.	• Inertia Formulas
- Calculation of Friction Torque Calculates the frictional force for each element, where necessary, and converts it to friction torque for a motor shaft. - Calculation of External Torque Calculates the external force for each element, where necessary, and converts it to external torque of a motor shaft. - Calculates the total load torque for the motor shaft conversion value.	• Load Torque Formulas
Select a motor temporarily based upon the motor shaft converted load inertia, friction torque, external torque and r.p.m. of a motor.	---
Calculate the Acceleration/Deceleration Torque from the Load Inertia or Operating Pattern.	• Acceleration/Deceleration Torque Formulas
- Calculate the necessary torque for each part of the Operating Pattern from the Friction Torque, External Torque and Acceleration/Deceleration Torque. - Confirm that the maximum value for the Torque for each operating part (Maximum Momentary Torque) is less than the Maximum Momentary Torque of the motor. - Calculate the Effective Torque from the Torque for each Operating part, and confirm that it is less than the Rated Torque for the motor.	• Calculation of Maximum Momentary Torque, Effective Torque



Explanation	References
Calculate Regenerative Energy from the Torque of all the moving parts.	Please see the user manual of each product for the details on calculation of the regenerative energy.
Check if the the number of encoder pulses meets the system specified resolution.	Accuracy of Positioning
Check if the calculation meets the specifications of the temporarily selected motor. If not, change the temporarily selected motor and re-calculate it.	The following table

Specialized Check Items	Check Items
Load Inertia	Load Inertia ≤ Motor Rotor Inertia x Applicable Inertia Ratio
Effective Torque	Effective Torque < Motor Rated Torque • Please allow a margin of about 20%. *
Maximum Momentary Torque	Maximum Momentary Torque < Motor Maximum Momentary Torque • Please allow a margin of about 20%. * • For the motor Maximum Momentary Torque, use the value that is combined with a driver and the one of the motor itself.
Maximum Rotation Speed	Maximum Rotation Speed ≤ Rated Rotation Speed of a motor • Try to get as close to the motor's rated rotations as possible. It will increase the operating efficiency of a motor. • For the formula, please see "Straight-line Speed and Motor Rotation Speed" on Page 16.
Regenerative Energy	Regenerative Energy ≤ Regenerative Energy Absorption of a motor • When the Regenerative Energy is large, connect a Regenerative Energy Absorption Resistance to increase the Absorption capacity of the driver.
Encoder Resolution	Ensure that the Encoder Resolution meets the system specifications.
Characteristics of a Positioner	Check if the Pulse Frequency does not exceed the Maximum Response Frequency or Maximum Command Frequency of a Positioner.
Operating Conditions	Ensure that values of the ambient operating temperature/humidity, operating atmosphere, shock and vibrations meet the product specifications.

* When handling vertical loads and a load affected by the external torque, allow for about 30% of capacity.

QUESTIONS ?